Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation

Florian Rzepka[a,*], Julia Kowal[a]

[a] *Technical University of Berlin, Chair of Electrical Energy Storage Technology, Einsteinufer 11, Berlin, 10587, Germany*

Abstract

State-of-charge (SOC) and state-of-health (SOH) estimators based on recurrent neural networks can achieve high accuracy, but their computational and memory cost on embedded hardware is often insufficiently characterised. This work quantifies the accuracy, timing, and resource trade-offs of Long Short-Term Memory (LSTM) and Multi-Layer Perceptron (MLP) based SOC/SOH estimators deployed on an STM32H753ZI microcontroller. Using the open *LFP DoE Cycle Ageing Dataset*, we train full-precision baseline models and obtain compressed variants via pruning and weight quantization. In long-horizon streaming replay on the test set the SOC estimators attain mean-absolute errors (MAE) between 2.3 % and 2.8 % of full scale, while the SOH estimators reach MAE values between 0.9 % and 1.5 %. Hardware-in-the-loop benchmarks on the STM32 show that pruning reduces both the flash footprint and the energy per inference by roughly 40 % for SOC and about 45 % for SOH, while slightly improving SOC MAE and keeping SOH MAE within a few tenths of a percent of the dense baseline. Quantization further shrinks the flash footprint to around half of the baseline size for both tasks and keeps the MAE within a few tenths of a percent of the corresponding full-precision models, at the expense of increased inference energy, especially for SOC. The presented methodology and measurements

*Corresponding author.
Email addresses: `florian.rzepka@tu-berlin.de` (Florian Rzepka), `julia.kowal@tu-berlin.de` (Julia Kowal)

provide a reproducible reference for implementing and comparing data-driven battery state estimators on resource-constrained Battery Management System (BMS) hardware.

Keywords: State of charge, State of health, LSTM, Pruning, Quantization, STM32, Battery diagnostics, Embedded AI

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CRediT authorship contribution statement

Florian Rzepka: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing–original draft preparation,

writing–review and editing, visualization. **Julia Kowal**: Resources, supervision, project administration, funding acquisition, writing–review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.